

Modeling Benzodiazepines on a Pyramidal Cell

A Complete, Step-by-Step Student Guide (NEURON simulator)

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Model: risecourse / BahlSynapsesPy

Read me first

Hi everyone! This guide walks you through the whole assignment slowly, one small step at a time. You do NOT need to be a programmer. By the end you will understand what a neuron is, what the code does, why each setting exists, and exactly what to change to model a benzodiazepine drug. Take it section by section -- nothing here is hard, I promise.

What's in this guide

1. The biology you need (neurons, GABA, benzodiazepines)
2. What is NEURON? What are .mod files?
3. The files in your folder
4. main.py explained line by line (every setting)
5. cellClasses.py explained (the cell and its synapses)
6. THE ASSIGNMENT: how to model a benzodiazepine
7. Installing and running it
8. Parameter cheat-sheet (what to change and why)
9. What you should see (results)
10. What to submit + troubleshooting
11. Glossary of every term

1. The Biology You Need

Before we touch any code, here is the whole story in plain language. If you understand this section, everything else is just details.

What is a neuron doing?

A neuron is a tiny cell that talks using electricity. It has a voltage across its membrane (like a tiny battery). When enough input pushes that voltage up past a line called the threshold, the neuron fires a spike (also called an action potential) -- a fast electrical pulse. Then it resets and can fire again. Spikes are the messages neurons send.

Two kinds of signals: gas and brakes

Excitatory signals push a neuron toward firing (the gas pedal). **Inhibitory** signals push it away from firing (the brake pedal). A healthy brain constantly balances the two.

GABA and the GABA_A receptor (the brake)

GABA is the brain's main *inhibitory* neurotransmitter -- the chemical the brakes are made of. When GABA lands on a **GABA_A receptor**, that receptor opens a channel that lets **chloride (Cl⁻)** ions into the cell. Chloride

makes the voltage go *down*, away from the firing threshold. So opening GABA_A channels = stronger braking
= the neuron fires less.

What a benzodiazepine actually does

A **benzodiazepine** (Valium, Xanax, Ativan...) is a calming/anti-anxiety drug. Here is the key idea most people get wrong: **the drug does nothing by itself**. It cannot inhibit a neuron on its own. Instead, it sits on the GABA_A receptor and makes it respond *more strongly* whenever GABA is present -- the channel opens more often, so each braking event is bigger and lasts longer.

The one sentence to remember

Benzodiazepines don't switch neurons off -- they turn UP the volume on the brakes the brain already uses (GABA_A inhibition). So, a firing pyramidal cell gets braked harder and fires LESS.

Our star: the pyramidal cell

The cell we simulate is a pyramidal cell -- the main excitatory neuron of the cortex. In the real brain it receives GABA_A inhibition from other cells. In our model we give it some current to make it fire, add GABA_A inhibition, then add a 'benzodiazepine' by making that inhibition stronger. We predict: the cell will fire fewer spikes. That is the entire experiment.

2. What is NEURON? What are .mod Files?

NEURON is a free simulator that scientists use to build realistic model neurons on a computer. You describe the cell's shape and its channels, press run, and NEURON solves the electricity equations for you.

Sections = the parts of the cell

In NEURON, a neuron is built from sections -- little cylinders of membrane. Our pyramidal cell has a soma (cell body), basal and apical dendrites, a tuft at the top, and an axon. NEURON knows the shape and the voltage -- but on its own it does NOT know what an ion channel is.

So, what teaches NEURON about channels? The .mod files

Each **.mod file** is a recipe for ONE membrane mechanism -- one type of ion channel, ion pump, or synapse. They are written in a small language called **NMODL** (NEURON MODEL language). Think of them as the *appliances* of the cell: the soma and dendrites are the rooms and wiring, and each .mod file is a device with its own behavior that you plug in.

Why do they have to be 'compiled'?

The channel math runs thousands of times per second for every part of the cell, so NEURON turns the .mod files into fast machine code first. That is what the command **nrnivmodl** does. You run it ONCE, and it builds the channels so the model can use them. (This is why the model won't run until you compile.)

The .mod files in YOUR folder

Two families. First, the ion channels and pumps that give the cell its electrical personality:

File	What it is	Its job
nat.mod	Fast sodium channel	Makes the spike shoot upward (the fast part of an action potential)
nap.mod	Persistent sodium	A small steady current that boosts excitability
kfast.mod	Fast potassium	Brings the spike back down quickly (repolarization)
kslow.mod	Slow potassium	Slower repolarization; controls firing rate
IKM.mod	M-type potassium (km)	Slow current that sets adaptation / how easily it fires
kca.mod	Calcium-activated K+	Links calcium level to firing (calms the cell after Ca enters)
SlowCa.mod	Slow calcium (sca)	Lets calcium into the top dendrites (tuft)
cad.mod	Calcium decay (cad)	Tracks and clears internal calcium (a pump/buffer)
h.mod	H-current (ih)	Hyperpolarization-activated current; shapes resting behavior

Second, the two 'point process' files --things that sit at ONE spot, not spread over the membrane:

File	What it is	Its job
my_exp2syn.mod	The synapse (MyExp2Syn)	The GABA_A (and AMPA) synapse --THIS is what the benzodiazepine acts on
mynetstim.mod	Spike generator (MyNetStim)	Fake 'input cells' that fire at a set rate and activate the synapses

Most important line to circle

my_exp2syn.mod is the GABA_A synapse -- the exact part a benzodiazepine grabs onto. You will NOT edit this file. Instead, you tune it with one setting in main.py (coming up in Section 6).

3. The Files in Your Folder

You don't need to open most of these. Here's the map so nothing feels mysterious:

File / folder	What it is
main.py	The control panel. You run this. All the settings you change live here.
cellClasses.py	Defines the pyramidal cell: its shape, channels, and synapses.
define_stimcells.py	Creates the artificial 'input cells' that send spikes to our neuron.
simrun.py	Helper that runs the simulation and draws the plots.
init_models_with_ca/	Presets that give the cell its exact electrical 'personality'.
*.mod files	The ion channels and synapses (Section 2). Compile once with nrnivmodl.
Results/	Where output plots and data files are saved after you run.
START_HERE.txt	Quick run instructions.

4. main.py Explained --Every Setting

main.py has a 'Set parameters' area near the top. These are the knobs you are allowed to turn. Let's go through each one slowly --what it is, and why it's there.

Temperature

```
h.celsius = 37 # degrees Celsius
```

Why: ion channels behave differently at different temperatures. 37°C is body temperature, so the model matches a living mammal. Leave it at 37.

Simulation name & plotting

```
simname = "testcell"    # a label for this run's output files
plotflag = 1            # 0 = no plots, 1 = show plots, 2 = show + save images
```

Why: simname just names the output files (so runs don't overwrite each other). plotflag decides whether pictures pop up. Use 1 while you work.

☆ The benzodiazepine knob (the important one)

```
myTauValue = 10         # GABA_A decay time in ms. 10 = control, raise it to add the drug
```

Why: this is the GABA_A synapse's **decay time** --how long each inhibitory (braking) event lasts. A benzodiazepine makes GABA_A inhibition last longer, so **raising this number = adding the drug**. We will use this in Section 6. (Behind the scenes it becomes the synapse's 'tau2' inside cellClasses.py.)

How long to simulate

```
mytstop = 800           # length of the simulation in milliseconds (ms)
```

Why: how much time to simulate. 800 ms is long enough to see plenty of spikes. (1000 ms = 1 second.)

What kind of input the cell gets

```
addSynInputs = 2        # 0 = current injection only
                        # 1 = synaptic inputs only
                        # 2 = BOTH current injection and synaptic inputs
```

Why: this chooses how we make the cell fire. We use **2**: a steady current makes the cell fire nicely, and the GABA_A synapses provide the inhibition our drug will strengthen. (With this setting the benzodiazepine effect is easiest to see.)

The current injection (used when addSynInputs is 0 or 2)

```
injectionLevel = 0.4     # nA  --how hard we push the cell (the 'gas')
injectionDuration = 500  # ms   --how long the push lasts
injectionStart = 100     # ms   --when the push begins
```

Why: these describe the electrode that drives the cell to fire. 0.4 nA is a gentle steady push, starting at 100 ms and lasting 500 ms. Bigger injectionLevel = more firing.

How fast the input cells fire

```
stimPeriod = 125         # ms between input spikes (125 ms = 8 times per second)
```

Why: the artificial input cells fire on a rhythm. 125 ms between spikes = an 8 Hz rhythm (like a brain wave). This drives both the excitatory and inhibitory synapses.

How many input cells

```
numExcCells = 2          # excitatory input cells
numInhDendCells = 2       # inhibitory input cells onto the dendrites
numInhSomaCells = 1       # inhibitory input cells onto the cell body
```

Why: these decide how many 'partner' neurons send signals to our cell. The inhibitory ones are the GABA_A senders --the ones the benzodiazepine boosts.

5. cellClasses.py --The Cell and Its Synapses

You do not need to edit this file, but you should understand two things inside it: how the cell is built, and where the GABA_A synapse is defined (because that's the drug's target).

Building the cell

The cell is assembled from sections and then channels are 'inserted' into different parts --for example the soma gets sodium and potassium channels so it can spike, the tuft gets calcium channels, and every part gets a passive leak. That mix is what makes it behave like a real layer-5 pyramidal cell.

☆ The GABA_A synapse (the benzodiazepine target)

Inside a method called addSynapses, the inhibitory synapses are created like this:

```
syn_ = h.MyExp2Syn(sec(0.5)) # make a synapse (from my_exp2syn.mod)
syn_.tau1 = 0.5              # RISE time: how fast it turns on (ms)
syn_.tau2 = myTauValue       # DECAY time: how long it lasts ← the drug!
syn_.e = -75                 # reversal potential (mV): makes it INHIBITORY
```

Read this carefully. **tau2** is set to **myTauValue** --the very knob from main.py. So when you raise myTauValue, every GABA_A synapse decays more slowly = stronger, longer inhibition = the benzodiazepine effect.

Why does e = -75 make it a 'brake'?

-75 mV is below the firing threshold, so opening this channel always pulls the voltage *down*, away from firing. That single number is what makes this synapse inhibitory. (The excitatory AMPA synapse in the same file uses e = 0 instead, which pulls the voltage UP.)

The synapse strengths (peak size)

```
excitatory_syn_weight = 0.005 # strength of excitatory (AMPA) synapses
inhDend_syn_weight    = 0.015 # strength of inhibitory synapses on dendrites
inhSoma_syn_weight    = 0.03  # strength of inhibitory synapses on the soma
```

Why: these set how big each synaptic event is. The two inhibitory ones are also GABA_A, so raising them is another way to strengthen inhibition. (In our model, changing the decay time --myTauValue --has a much bigger effect than these, which is why that's the knob we use.)

6. ☆ THE ASSIGNMENT: Model the Benzodiazepine

Here is the actual task, and it is refreshingly simple.

What to do

In main.py, change **myTauValue** to a larger number to model the drug, then run the model and see that the pyramidal cell fires fewer spikes.

The logic, step by step:

1. A benzodiazepine strengthens and prolongs GABA_A inhibition (Section 1).

2. In this model, GABA_A inhibition length = the synapse decay time = myTauValue (Section 5).
3. So, to add the drug, we RAISE myTauValue.
4. Stronger inhibition = the brakes win more = the cell fires FEWER spikes.

Suggested values to try:

Set myTauValue to	This represents	Roughly this many spikes
10	Control (no drug)	about 10 spikes
40	A benzodiazepine	about 7 spikes
100	A strong dose	about 5 spikes

That decreasing pattern (10 → 7 → 5) IS the benzodiazepine effect. Notice the drug never touches the 'gas' (the current injection). It only strengthens the brakes.

7. Installing and Running It

First: install NEURON (one-time setup)

You need the NEURON simulator plus a couple of Python packages. If your RISE Anaconda environment already has NEURON, skip ahead to “Then: run the model.” Otherwise:

5. Open the Anaconda Prompt (Windows) or a Terminal (Mac/Linux), and turn on your RISE environment (use the name your instructor gives you):

```
conda activate rise      # replace 'rise' with your environment name
```

6. Go into this lab folder, then install the packages:

```
pip install -r requirements.txt      # Mac / Linux
```

Windows is different

On Windows, **pip install neuron does NOT work**. Instead, download the NEURON installer (.exe) from github.com/neuronsimulator/nrn/releases, run it like a normal program, then do: `pip install numpy matplotlib`. Full details are in INSTALL.txt in this folder.

7. Check it worked --this should print a version number:

```
python -c "import neuron; from neuron import h; print(h.nrnversion())"
```

Then: run the model

8. Compile the channels ONCE (this reads all the .mod files):

```
nrnivmodl      # on Windows use: mknrndll
```

9. Make sure main.py has myTauValue = 10 for your first (control) run.

10. Run the model:

```
python main.py      # or press Run in Spyder
```

11. Look in the Results folder --you'll see plots of the voltage and a list of spike times. Count or note the spikes.
12. Now change myTauValue to 40 (the benzodiazepine), run again, and compare. Fewer spikes = success! Then try 100.

If Spyder can't find NEURON

Close Spyder, activate your RISE environment in a terminal first, then launch Spyder from there (type spyder). Spyder must run inside the same environment where NEURON is installed.

8. Parameter Cheat-Sheet

A quick reference of every knob and what happens when you change it:

Parameter	What it controls	Try changing it to...
myTauValue	GABA _A inhibition length (THE drug knob)	10 (control) → 40 → 100 to add more drug
addSynInputs	Type of input (0/1/2)	Keep at 2 for this assignment
injectionLevel	Strength of the 'gas' current (nA)	Raise to 0.5 for more firing; lower for less
mytstop	Simulation length (ms)	Longer to see more spikes (e.g. 1000)
stimPeriod	Rhythm of input cells (ms)	Lower = faster input rhythm
plotflag	Show/save plots	1 while working, 2 to also save images

9. What You Should See

When you raise myTauValue, the spikes should thin out. You should see roughly:

- Control (myTauValue = 10): about 10 spikes --the cell fires freely.
- Benzodiazepine (myTauValue = 40): about 7 spikes --noticeably fewer.
- Strong dose (myTauValue = 100): about 5 spikes --clearly suppressed.

If your numbers are close to these, you did it right. The exact counts can vary a little, but the direction must be the same: more drug → fewer spikes. Compare your own control run and your drug run to see the difference.

10. What to Submit + Troubleshooting

Submit to Blackboard

- main.py (you changed one line: myTauValue)
- cellClasses.py (submit it unchanged --it already sends myTauValue to the synapse)

Do NOT submit the Results files or the compiled channel folder.

Troubleshooting

“nrnivmodl fails on mynetstim.mod” with an error about 'nrn_random_arg': this happens on newer NEURON (version 9). Fix: open mynetstim.mod, find these two lines near line 90 and delete them:

```
double nrn_random_pick(void* r);  
void* nrn_random_arg(int argpos);
```

Then run nrnivmodl again. (In the folder I gave you, this is already fixed.)

“The cell never fires at all.” Make sure addSynInputs = 2 and injectionLevel is around 0.4. If it still won't fire, raise injectionLevel a little.

“Changing myTauValue does nothing.” Make sure you saved main.py before running, and that you recompiled with nrnivmodl at least once.

11. Glossary --Every Term in One Place

Term	Plain-English meaning
Neuron	A brain cell that signals using electricity.
Spike / action potential	A fast electrical pulse a neuron fires to send a message.
Threshold	The voltage line a neuron must cross to fire a spike.
Excitatory	A signal that pushes a neuron toward firing (the gas).
Inhibitory	A signal that pushes a neuron away from firing (the brake).
GABA	The brain's main inhibitory (braking) neurotransmitter.
GABA_A receptor	The channel GABA opens; lets chloride in and lowers voltage.
Benzodiazepine	A drug that strengthens GABA_A inhibition (only when GABA is present).
Pyramidal cell	The main excitatory neuron of the cortex --our model cell.
NEURON	The software we use to simulate neurons.
.mod file	A recipe for one ion channel, pump, or synapse (in NMODL language).
nrnivmodl	The command that compiles all .mod files so the model can use them.
Section	A piece of the cell in NEURON (soma, dendrite, etc.).
Synapse	The connection where one neuron's signal reaches another.
tau1 / tau2	How fast a synapse turns on (rise) / how long it lasts (decay).
Reversal potential (e)	The voltage a synapse pulls toward; -75 = inhibitory, 0 = excitatory.
myTauValue	Our knob for GABA_A decay time --raising it models the drug.

You've got this. Read it once slowly, run the control, then add the drug. --Sai